

Random Variables, X

↳ A function that assigns a numerical value to each outcome in a sample space of a random experiment.

↳ Types of Random Variables:



Discrete

↳ Countable number of distinct values

↳ Example: # of heads in n coin flips

Continuous

↳ Uncountable numbers within an interval

↳ Example: heights of people within six feet

◆ 1. The Random Variable is the Observation

- This is the simplest relationship.
- The outcome of a random experiment is directly recorded as the value of the random variable.

✓ Example:

You toss a coin:

- Observation: "Head"
- Define random variable X such that:
 - $X = 1$ if Head
 - $X = 0$ if Tail

So the random variable directly represents the observed outcome.

◆ 3. The Random Variable is a Function of Another Random Variable

- You define one random variable based on another random variable.
- This is common in transformations, derived distributions, or conditional probability models.

✓ Example:

Let X = number of heads in 3 coin tosses.

Define $Y = 2X + 1$

- Here, Y is a function of another random variable X .
- If $X = 2$ (i.e., 2 heads), then $Y = 2(2) + 1 = 5$

◆ 2. The Random Variable is a Function of the Observation

- The observation is more complex, and the random variable is derived by applying a function to that observation.

✓ Example:

You roll a pair of dice:

- Observation: (3, 4)
- Random variable X = sum of dice = $3 + 4 = 7$

Here, the random variable is a function of the observation:

$$X = f(\text{observation}) = f(3, 4) = 7$$

✦ Summary Table:

Type	Description	Example
1. Random variable is the observation	Direct mapping	Toss coin $\rightarrow X = 1$ (Head), 0 (Tail)
2. Function of observation	Derived from raw data	Dice roll (2,5) $\rightarrow X = \text{sum} = 7$
3. Function of another random variable	Transformed variable	$X = \text{heads in tosses}$, $Y = 2X + 1$



In summary, a random variable is a numerical representation of outcomes in a random experiment.

Basic Example: Flip a coin 4 times

∴ $\Omega = \{ \text{HHHH}, \text{HHHT}, \text{HHTT}, \dots, \text{T T T T} \}$

if, $X = \#$ heads that occur.

$$X \in \{0, 1, 2, 3, 4\}$$

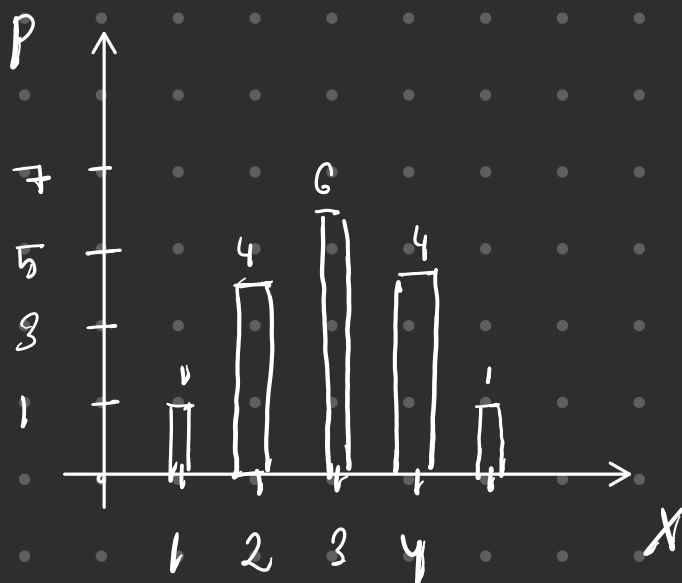
$$P(X=0) = \frac{1}{16}$$

$$P(X=1) = \frac{4}{16}$$

$$P(X=2) = \frac{6}{16}$$

$$P(X=3) = \frac{4}{16}$$

$$P(X=4) = \frac{1}{16}$$



So, the formal definition of a random variable $P(X)$ is a function called a distribution that contains

↳ values of X

↳ Probability P of each value

✱ It gives a concise summary of all probabilities and helps to visualize and compute.

PMF - Probability Mass Function

↳ Used in Discrete Random Variables

↳ Gives the probability that a discrete random variable is exactly equal to a value.

⊗ If X is a discrete random variable, then PMS is $P(X=x)$
This gives the probability that the random variable equals x .

↳ For example: Toss a fair coin once:

Let X = number of heads. | ⊗ $P(X=1) = 0.5$
⊗ $P(X=0) = 0.5$

PDF - Probability Density Function (will be discussed in next chapter)

↳ Used in Continuous Random Variables.

↳ Gives the density of the probability at a point.
It doesn't give the actual probability of a value.
The probability is computed over an interval using integration

⊗ If X is a continuous random variable:
$$P(a \leq X \leq b) = \int_a^b f(x) dx \quad ; \quad f(x) \rightarrow \text{PDF}$$

↳ For example: Let $X \sim U(0,1)$ (uniform distribution from 0 to 1) then: o.c
 $f(x) = 1$ for $0 \leq x \leq 1$ || $P(0.2 \leq x \leq 0.6) = \int_{0.2}^{0.6} 1 dx = 0.4$

* For a discrete random variable X with PMF $P_X(x)$ and range S_X .

(a) For any x , $P_X(x) \geq 0$ [Non-negativity]

(b) $\sum_{x \in S_X} P(x) = 1$. [Normalization]

(c) For any event $B \subset S_X$, the probability that X is in the set B is $P[B] = \sum_{x \in B} P_X(x)$ [Additivity for events]

The random variable N has PMF

$$P_N(n) = \begin{cases} c/n & n = 1, 2, 3, \\ 0 & \text{otherwise.} \end{cases}$$

Find

(a) The value of the constant c

(b) $P[N = 1]$

(c) $P[N \geq 2]$

(d) $P[N > 3]$

Example:

Solⁿ:

(a) $P_N(n)$ is a PMF.

$$\text{Here, } \sum_{n \in S_N} P(n) = 1$$

$$\rightarrow \sum_{n=1}^3 P_N(n) = 1 \Rightarrow \frac{c}{1} + \frac{c}{2} + \frac{c}{3} = 1 \Rightarrow C = \frac{6}{11}$$

$$(b) \text{ If, } P[N=1] = \frac{C}{1} = \frac{6}{11} \\ \therefore P[N=1] = \frac{6}{11}$$

$N \rightarrow$ random variable
 $n \rightarrow$ specific value N takes.

$$\textcircled{c} \quad P[N \geq 2] = P[N=2] + P[N=3] = \frac{C}{2} + \frac{C}{3}$$

$$= \frac{6}{11} \left(\frac{1}{2} + \frac{1}{3} \right)$$

$$= \frac{5}{11}$$

$\textcircled{d} \quad P[N > 3] = 0$; PMF is defined only for $n = 1, 2, 3, 0$

Example from Slide:

Example 2.1: (Continued)

Procedure: Send 3 packets from a sender to a receiver.

Observation: Number of successes.

$S = \{FFF, FFD, FDF, FDD, DFF, DFD, DDF, DDD\}$

$E = \{E_0, E_1, E_2, E_3\}$

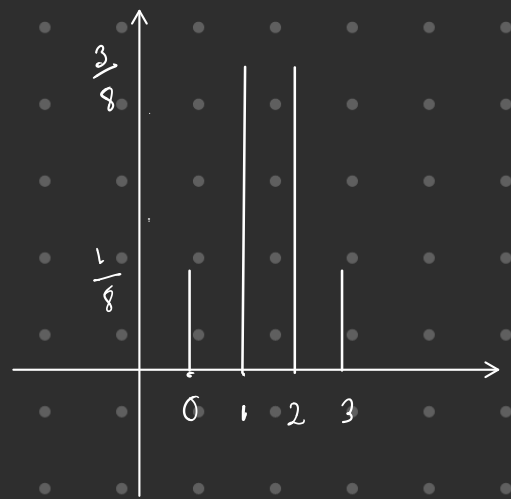
$X \triangleq$ Random variable that counts the number of successes

$S_X = \{0, 1, 2, 3\}$

n (successes)	E_x	s (sequences)	$P[X=n]$	$P[X \leq n]$
0	E_0	DDD	$1/8$	$1/8$
1	E_1	FDD, DFD, DDF	$3/8$	$1/8 + 3/8$
2	E_2	FFD, FDF, DFF	$3/8$	$1/8 + 3/8 + 3/8$
3	E_3	FFF	$1/8$	$1/8 + 3/8 + 3/8 + 1/8$

So, the PMF would be:

$$P_X(x) = \begin{cases} 1/8 & \text{if } x = 0, 3 \\ 3/8 & \text{if } x = 1, 2 \\ 0 & \text{otherwise} \end{cases}$$



⊗ Falls onto Binomial Distribution Technique.
This will be shown later on.

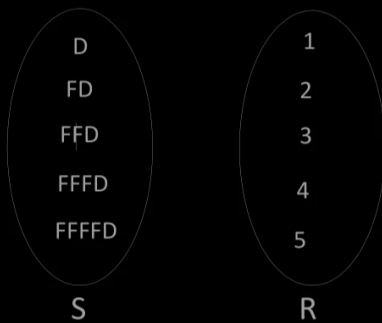
Example from Slides

Example 2.2 (Example 1.13 Continued)

Procedure: Keep sending packets from a sender to a receiver until 1 packet is delivered

Observation: Number of attempts

$$S = \{D, FD, FFD, FFFD, \dots\} \quad s_X = \{1, 2, 3, \dots\}$$



X = number of attempts until success.

Possible values: $s_X = \{1, 2, 3, \dots\}$

$$p(\text{success}) = p = 0.5$$

$$p(\text{failure}) = 1 - p = 0.5$$

	E_1	E_2	E_3	E_4	E_5	E_6
S	D	FD	FFD	FFFD	FFFFD	FFFFFD
x	1	2	3	4	5	6
$P[X=x]$	0.5	0.5^2	0.5^3	0.5^4	0.5^5	0.5^6
$P[X \leq x]$	0.5	$0.5 + 0.5^2$	$0.5 + 0.5^2 + 0.5^3$	$0.5 + 0.5^2 + 0.5^3 + 0.5^4$	$0.5 + 0.5^2 + 0.5^3 + 0.5^4 + 0.5^5$	$0.5 + 0.5^2 + 0.5^3 + 0.5^4 + 0.5^5 + 0.5^6$

∴ PMF : $P(X=x) = \begin{cases} (0.5)^x & \text{for } x = 1, 2, 3, \dots \\ 0 & \text{otherwise} \end{cases}$

⊛ Falls onto Geometric Distribution Technique.
This will be shown later on.

Example

from Slide:

X = number of packets until three successes

$$P(\text{success}) = p = 0.5$$

$r = 3$ (number of success we want)

Example 2.3: (Example 1.14 Continued)

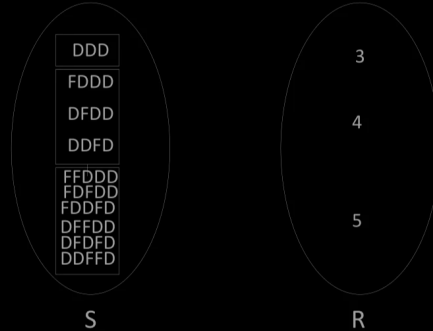
Example 1.14 (Experiment 1.5)

Procedure: Keep sending packets from a sender to a receiver until 3 packets are delivered

Observation: Number of attempts

$$S = \{DDD, FDDD, DFDD, DDFD, FFDDD, \dots\}$$

$$S_X = \{3, 4, 5, \dots\}$$



S

x

$$P[X=x]$$

$$P[X \leq x]$$

E_3

DDD

3

$$p^3$$

$$p^3$$

E_4

FDDD, DFDD,
DDFD

4

$$3 \cdot p^3(1-p)$$

$$3p^3(1-p) + p^3$$

E_5

FFDDD, FDFDD,
FDDFD, DFFDD,
DDFDD, DDFFD.

5

$$6 \cdot p^3(1-p)^2$$

$$p^3 + 3p^3(1-p) + 6p^3(1-p)^2$$

$$\therefore \text{PMF} : P(X=x) = \begin{cases} \binom{x-1}{r-1} p^r (1-p)^{x-r}, & x=3,4,5,\dots \\ 0 & \text{otherwise} \end{cases}$$

* Falls onto Pascal Distribution Technique
This will be shown later on.

CDF - Cumulative Distribution Function

↳ denoted by $F_X(x)$, gives the probability that X takes a value less than or equal to x ,

$$F_X(x) = P(X \leq x)$$

This accumulates the probabilities of all outcomes up to and including x . So, for any real number x , the CDF is the probability that the random variable X is no larger than x .

If X is discrete, the CDF is the sum of the PMF values up to x :

$$F_X(x) = \sum_{k=-\infty}^{\lfloor x \rfloor} p_X(k)$$

⊗ All random variables have cumulative distribution functions but only discrete random variables have probability mass functions.

Properties of CDF:

① Non-decreasing: CDF always increases or stays constant. For all $x' \geq x$, $F_X(x') \geq F_X(x)$.

② $\lim_{x \rightarrow -\infty} F_X(x) = 0$ and $\lim_{x \rightarrow \infty} F_X(x) = 1$

Can be used to compute probability ranges:

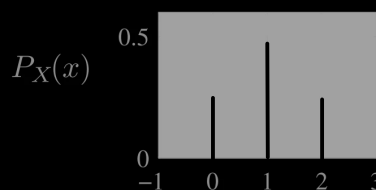
$$P(a < X \leq b) = F_X(b) - F_X(a)$$

(iv) $F_X(x) = F_X(x_i)$ for all x such that $x_i \leq x < x_{i+1}$

Example:

Example 3.21

In Example 3.5, random variable X has PMF



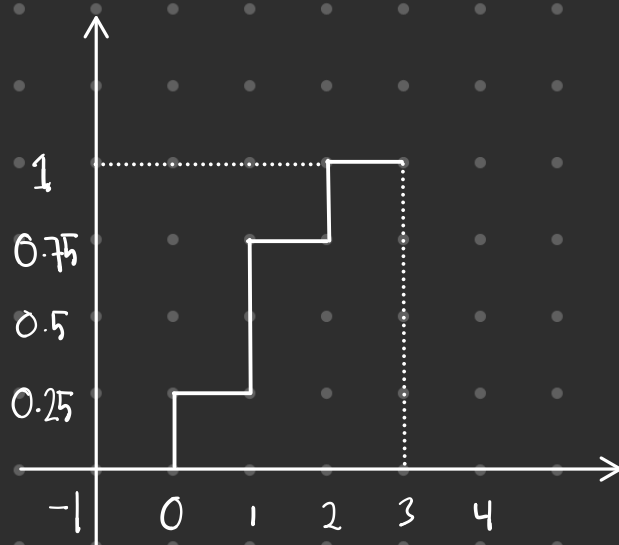
$$P_X(x) = \begin{cases} 1/4 & x = 0, \\ 1/2 & x = 1, \\ 1/4 & x = 2, \\ 0 & \text{otherwise.} \end{cases} \quad (3.28)$$

Find and sketch the CDF of random variable X .

Soln:

$$F_X(x) = P[X \leq x]$$

$$= \begin{cases} 0 & x < 0 \\ 1/4 & 0 \leq x < 1 \\ 3/4 & 1 \leq x < 2 \\ 1 & x \geq 2 \end{cases}$$



Example from the slide:

Example 2.4: Let a discrete random variable X assumes values $-1, 1, 2$, and 3 , with probabilities $0.6, 0.3, 0.08$, and 0.02 , respectively.

$$\text{PMF: } P_X(x) = \begin{cases} 0.6, & x = -1 \\ 0.3, & x = 1 \\ 0.08, & x = 2 \\ 0.02, & x = 3 \\ 0, & \text{otherwise} \end{cases}$$

CDF:

$$F_X(x) = \begin{cases} 0, & x \leq -1 \\ 0.6, & -1 \leq x < 1 \\ 0.9, & 1 \leq x < 2 \\ 0.98, & 2 \leq x < 3 \\ 1, & x \geq 3 \end{cases}$$

Bernoulli (p) Random Variables.

↳ Models an experiment that has only two possible outcomes typically success (1) or failure (0). A Bernoulli random variable X is defined as:

$$X = \begin{cases} 1, & \text{with probability } p \\ 0, & \text{with probability } 1-p \end{cases} \quad \left| \begin{array}{l} p \in [0,1] \rightarrow \text{success} \\ 1-p \rightarrow \text{failure} \end{array} \right.$$

Bernoulli PMF:

$$P_X(x) = \begin{cases} p & \text{if } x=1 \\ 1-p & \text{if } x=0 \\ 0 & \text{otherwise} \end{cases}$$

For example:

A biased coin:

$$P[\text{Heads}] = 0.7$$

$$P[\text{Tails}] = 0.3$$

$$\text{So, } X = \begin{cases} 1, & \text{if heads} \\ 0, & \text{if tails} \end{cases}$$

$$\therefore P_X(x) = \begin{cases} 0.7 & \text{if } x=1 \\ 0.3 & \text{if } x=0 \\ 0 & \text{otherwise} \end{cases}$$

Geometric Random Variable

↳ A geometric random variable models the number of trials until the first success in a sequence of independent Bernoulli trials (each with success- p)

if, X = the number of trials until the first success

$$P(X=k) = (1-p)^{k-1} \cdot p \quad \text{for } k=1, 2, 3, \dots$$

$$\text{Geometric PMF, } P_X(x) = \begin{cases} p(1-p)^{k-1} & k=1, 2, 3, \dots \\ 0 & \text{otherwise} \end{cases}$$

Example: →

In a sequence of independent tests of integrated circuits, each circuit is rejected with probability p . Let Y equal the number of tests up to and including the first test that results in a reject. What is the PMF of Y ?

Solⁿ: Here,

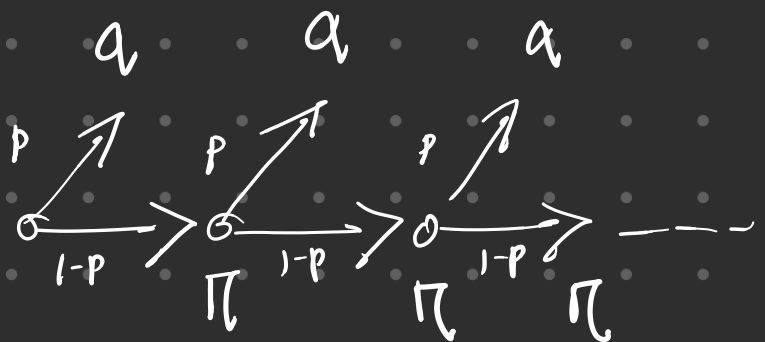
$$P[Y=1] = p$$

$$P[Y=2] = p(1-p)$$

$$P[Y=3] = p(1-p)^2$$

⋮

$$P[Y=k] = p(1-p)^{k-1}$$



$$P_Y(y) = \begin{cases} p(1-p)^{y-1} & y=1, 2, \dots \\ 0 & \text{otherwise} \end{cases}$$

Binomial Random Variable (n, p)

↳ Models the number of successes in a fixed number of independent trials, where each trial has only two outcomes: success (p) or failure ($1-p$)

Let: $n \rightarrow$ number of trial

$p \rightarrow$ probability of success in each trial

$X \rightarrow$ number of successes in those n trials

Then, $X \sim \text{Binomial}(n, p)$

So, the PMF: the probability of getting exactly k successes in n trials is:

$$P(X=k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad k = 0, 1, 2, \dots, n$$

For Example: Flip a biased coin 5 times where the probability of heads is $p = 0.6$

Let X be the number of heads in those 5 flips.

$$P(X=3) = \binom{5}{3} \cdot 0.6^3 \cdot 0.4^2 = 0.3456.$$

Recapping this example?

Example 2.1: (Continued)

Procedure: Send 3 packets from a sender to a receiver.

Observation: Number of successes.

$$S = \{FFF, FFD, FDF, FDD, DFF, DFD, DDF, DDD\}$$

$$E = \{E_0, E_1, E_2, E_3\}$$

$X \triangleq$ Random variable that counts the number of successes

$$S_X = \{0, 1, 2, 3\}$$

Using Binomial Distribution:

$$P_X(x) = \binom{3}{x} \left(\frac{1}{2}\right)^x \left(1 - \frac{1}{2}\right)^{3-x}$$

for $x = 0, 1, 2, 3$

$$P(X=0) = \frac{1}{8}, \quad P(X=1) = \frac{3}{8}$$

$$P(X=2) = \frac{3}{8}, \quad P(X=3) = \frac{1}{8}$$

Example 3.12: If there is a 0.2 probability of a reject and we perform 10 tests, the PMF of the binomial (10, 0.2) random variable is what?

Solⁿ: $P_K(k) = \binom{10}{k} (0.2)^k (0.8)^{10-k}$

Example 3.13: Perform independent tests of integrated circuits in which each circuit is rejected with probability p . Observe L , the number of tests performed until there are k rejects. What is the PMF of L ?

Solⁿ: Assuming, $L=l$ means the l -th test is the k -th rejection. This happens if:

→ The first $l-1$ tests have exactly $k-1$ rejections.

→ The l -th test is a rejection.

So, the PMF would be:

$$P(L=l) = \binom{l-1}{k-1} p^k (1-p)^{l-k} \text{ for } l \geq k$$

For Example:

Rejections needed, $k=2$

Rejection rate, $p=0.1$

Tests until the second rejection, $l=5$.

$$\therefore P(L=5) = \binom{5-1}{2-1} (0.1)^2 (1-0.1)^{5-2} = 0.02916$$

→ Pascal Random Variable

Pascal Random Variable (k, p)

A Pascal (k, p) random variable X represents the number of Bernoulli trials needed to achieve k successes, where each trial has success probability p .

↳ You are repeating Bernoulli trials until you get k successes.

↳ x is the total number of trials needed to get these k successes.

So, the PMF would be:

$$P_X(x) = \binom{x-1}{k-1} p^k (1-p)^{x-k}$$

$$x = k, k+1, k+2, \dots$$

$$p \in (0, 1)$$

$$k \in \mathbb{Z}^+ \text{ (positive integers)}$$

* If $k=1$, the Pascal distribution becomes a Geometric distribution (which models the number of trials until the first success)

$$\text{So, Pascal}(1, p) = \text{Geometric}(p)$$

For Example:

Flip a biased coin where the probability of getting heads $p = 0.6$. What's the probability to get the 3rd Head on the 5th flip?

Solⁿ:

$$P(X=5) = \binom{5-1}{3-1} (0.6)^3 (0.4)^{5-3} = 0.20736$$

Example 3.14 If there is a 0.2 probability of a reject and we seek four defective circuits, the random variable L is the number of tests necessary to find the four circuits. The PMF of the $\text{Pascal}(4, 0.2)$ random variable is what?

Solⁿ: $k=4, p=0.2.$

$$P_L(l) = \binom{l-1}{3} (0.2)^3 (0.8)^{l-4}$$

Discrete Uniform (k, l) Random Variable:

A discrete Uniform Random Variable X is one where each value between k and l (inclusive) is equally likely. The PMF:

$$P_N(n) = \begin{cases} \frac{1}{l-k+1} & \text{if } n = k, k+1, \dots, l \\ 0 & \text{otherwise} \end{cases}$$

⚡ All values in the range have equal probability. $k < l$, and both must be integers.

Example: Roll a fair six-sided die. Let N be the number of spots on the face top. Find PMF.

Solⁿ:

$$P_N(n) = \begin{cases} \frac{1}{6}, & \text{for } n = 1, 2, 3, 4, 5, 6 \\ 0 & \text{otherwise} \end{cases}$$

